

Kasimsetty Venkata Naga Arjun

arjun.kondala2005@gmail.com | +91 99897 43999 | [GitHub](#) | [LinkedIn](#)

EXPERIENCE

Skyeris Aerotech

Lead Designer & Systems Integrator

Remote

Jan 2025 – Oct 2025

- Designed F405-based modular flight controller supporting ArduPilot, INAV, and Betaflight
- Architected embedded systems with optimized interrupt handling and DMA for real-time performance

IISER Bhopal

Research Intern

Bhopal, India

May 2025 – Aug 2025

- Developed GPS-denied visual navigation system using deep learning and feature-based localization
- Built automated data pipelines for training and evaluation of autonomous systems

AP Police Department

Embedded Systems Engineer

India

May 2024 – Jul 2025

- Developed LoRa-WAN GPS tracker for long-range asset tracking
- Integrated real-time visualization using mapping APIs

PROJECTS

FlexML (Swift Student Challenge Winner) | *Developer*

- Built iOS system converting smartphones into real-time vision + telemetry nodes for robotics
- Enabled dynamic Core ML model deployment with on-device inference (30 FPS)
- Integrated ESP32-based systems via low-latency network streaming
- Optimized AVFoundation buffer pipelines for real-time performance

GPS-Denied Drone Navigation System | *Research & Development*

- Developed NetVLAD-based visual localization using preprocessed satellite imagery
- Achieved 99% accuracy and 95% precision in simulation (AirSim + Unreal Engine)
- Built preprocessing, training, and path planning pipeline in Python

AI-Based Water Quality Assessment | *Developer*

- Designed non-contact system to classify water potability using RGB imaging
- Built dataset from real samples (Vijayawada, Amaravathi) with lab validation
- Implemented preprocessing and inference pipeline with data augmentation

Tethered Drone Power System | *Hardware Designer*

- Designed PCB integrating TDK DC-DC modules for tethered drone power delivery
- Achieved 90% efficiency under weight-constrained aerial conditions
- Conducted LTSpice simulations and real-world validation

LEADERSHIP & EXTRACURRICULARS

Team TEDH | *Founder & Technical Lead*

- Led 5-member team to win multiple national-level hackathons (4× wins)
- Built full-stack engineering solutions combining hardware, software, and AI
- Managed task allocation, system design, and rapid prototyping workflows

Music Club, SRM University | *Social Media Head (Co-Convener)*

- Grew social media presence from 300 to 1000+ followers
- Achieved 300,000 monthly views through content strategy
- Built brand identity and improved engagement across events

TECHNICAL SKILLS

- **Embedded Systems:** STM32, ESP32, nRF52, FreeRTOS
- **Programming:** C, Python, Swift, Kotlin, Embedded C
- **Hardware:** KiCad, EasyEDA, CST, LT-Spice, Multisim
- **Protocols:** LoRaWAN, SPI, I2C, UART, CAN, MQTT, USB
- **AI/ML:** PyTorch, TensorFlow Lite, OpenCV
- **Tools:** Git, Oscilloscope, Logic Analyzer, JTAG

ACHIEVEMENTS & AWARDS

- Winner – Swift Student Challenge 2026 (Apple)
- Finalist – Smart India Hackathon 2025
- 4 × National-Level Hackathon Winner

EDUCATION

SRM University Amaravathi

B.Tech in Electronics and Communication Engineering

Minor in Computer Science

GPA: 8.19/10.0